

Optimisation Report

Before Index:

```
[
  {
    "$clusterTime": {
      "clusterTime": {"$timestamp": {"t": 1782210502, "i": 1}},
      "signature": {
        "hash": {"$binary": {"base64": "eFpW2I2WQiFdD0c2+q2IfTM7pqM
=", "subType": "00"}},
        "keyId": 7619114555372732417
      }
    },
    "command": {
      "find": "books",
      "filter": {
        "category": "Programming",
        "published_year": {
          "$gte": 2020
        }
      },
      "$db": "test"
    },
    "executionStats": {
      "executionSuccess": true,
      "nReturned": 4,
      "executionTimeMillis": 0,
      "totalKeysExamined": 0,
      "totalDocsExamined": 33,
      "executionStages": {
        "isCached": false,
        "stage": "COLLSCAN",
```

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"filter": {
  "$and": [
    {
      "category": {
        "$eq": "Programming"
      }
    },
    {
      "published_year": {
        "$gte": 2020
      }
    }
  ],
  "nReturned": 4,
  "executionTimeMillisEstimate": 0,
  "works": 34,
  "advanced": 4,
  "needTime": 29,
  "needYield": 0,
  "saveState": 0,
  "restoreState": 0,
  "isEOF": 1,
  "direction": "forward",
  "docsExamined": 33
},
"explainVersion": "1",
"ok": 1,
"operationTime": {"$timestamp": {"t": 1782210502, "i": 1}},
"queryPlanner": {
  "namespace": "test.books",
  "parsedQuery": {
    "$and": [
      {
        "category": {

```

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"$eq": "Programming"
}
},
{
  "published_year": {
    "$gte": 2020
  }
}
]
},
"indexFilterSet": false,
"queryHash": "92EDD004",
"planCacheShapeHash": "92EDD004",
"planCacheKey": "195A970C",
"optimizationTimeMillis": 0,
"maxIndexedOrSolutionsReached": false,
"maxIndexedAndSolutionsReached": false,
"maxScansToExplodeReached": false,
"prunedSimilarIndexes": false,
"winningPlan": {
  "isCached": false,
  "stage": "COLLSCAN",
  "filter": {
    "$and": [
      {
        "category": {
          "$eq": "Programming"
        }
      }
    ],
    {
      "published_year": {
        "$gte": 2020
      }
    }
  ]
}
},

```

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"direction": "forward"
},
"rejectedPlans": []
},
"queryShapeHash": "AA3078A59481B28A38775FA280FC4779BEA0E8E900
AF63A4344B76A85859C985",
"serverInfo": {
"host": "ac-8kf2wgk-shard-00-01.okhgovl.mongodb.net",
"port": 27017,
"version": "8.0.26",
"gitVersion": "a2c24805265db119cc39f2f54e067b4294a4ddd2"
},
"serverParameters": {
"internalQueryFacetBufferSizeBytes": 104857600,
"internalQueryFacetMaxOutputDocSizeBytes": 104857600,
"internalLookupStageIntermediateDocumentMaxSizeBytes": 167936
00,
"internalDocumentSourceGroupMaxMemoryBytes": 104857600,
"internalQueryMaxBlockingSortMemoryUsageBytes": 33554432,
"internalQueryProhibitBlockingMergeOnMongoS": 0,
"internalQueryMaxAddToSetBytes": 104857600,
"internalDocumentSourceSetWindowFieldsMaxMemoryBytes": 104857
600,
"internalQueryFrameworkControl": "trySbeRestricted",
"internalQueryPlannerIgnoreIndexWithCollationForRegex": 1
}
}
]

```

After Index:

```

[
{
"$clusterTime": {

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```

"clusterTime": {"$timestamp": {"t": 1782210552, "i": 2}},
"signature": {
  "hash": {"$binary": {"base64": "StWrHBcnm0K0JqTsgwZ9ztkcURU
=", "subType": "00"}},
"keyId": 7619114555372732417
}
},
"command": {
  "find": "books",
  "filter": {
    "category": "Programming",
    "published_year": {
      "$gte": 2020
    }
  },
  "$db": "test"
},
"executionStats": {
  "executionSuccess": true,
  "nReturned": 4,
  "executionTimeMillis": 1,
  "totalKeysExamined": 4,
  "totalDocsExamined": 4,
  "executionStages": {
    "isCached": false,
    "stage": "FETCH",
    "nReturned": 4,
    "executionTimeMillisEstimate": 0,
    "works": 5,
    "advanced": 4,
    "needTime": 0,
    "needYield": 0,
    "saveState": 0,
    "restoreState": 0,
    "isEOF": 1,
    "docsExamined": 4,

```

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"alreadyHasObj": 0,
"inputStage": {
  "stage": "IXSCAN",
  "nReturned": 4,
  "executionTimeMillisEstimate": 0,
  "works": 5,
  "advanced": 4,
  "needTime": 0,
  "needYield": 0,
  "saveState": 0,
  "restoreState": 0,
  "isEOF": 1,
  "keyPattern": {
    "category": 1,
    "published_year": 1
  },
  "indexName": "category_1_published_year_1",
  "isMultiKey": false,
  "multiKeyPaths": {
    "category": [],
    "published_year": []
  },
  "isUnique": false,
  "isSparse": false,
  "isPartial": false,
  "indexVersion": 2,
  "direction": "forward",
  "indexBounds": {
    "category": ["["Programming", "Programming"]"],
    "published_year": ["[2020, inf.0]"]
  },
  "keysExamined": 4,
  "seeks": 1,
  "dupsTested": 0,
  "dupsDropped": 0
}

```

```

}
},
"explainVersion": "1",
"ok": 1,
"operationTime": {"$timestamp": {"t": 1782210552, "i": 2}},
"queryPlanner": {
  "namespace": "test.books",
  "parsedQuery": {
    "$and": [
      {
        "category": {
          "$eq": "Programming"
        }
      },
      {
        "published_year": {
          "$gte": 2020
        }
      }
    ]
  },
  "indexFilterSet": false,
  "queryHash": "92EDD004",
  "planCacheShapeHash": "92EDD004",
  "planCacheKey": "795A952C",
  "optimizationTimeMillis": 0,
  "maxIndexedOrSolutionsReached": false,
  "maxIndexedAndSolutionsReached": false,
  "maxScansToExplodeReached": false,
  "prunedSimilarIndexes": false,
  "winningPlan": {
    "isCached": false,
    "stage": "FETCH",
    "inputStage": {
      "stage": "IXSCAN",
      "keyPattern": {

```

```

"category": 1,
"published_year": 1
},
"indexName": "category_1_published_year_1",
"isMultiKey": false,
"multiKeyPaths": {
"category": [],
"published_year": []
},
"isUnique": false,
"isSparse": false,
"isPartial": false,
"indexVersion": 2,
"direction": "forward",
"indexBounds": {
"category": ["[\"Programming\", \"Programming\"]"],
"published_year": ["[2020, inf.0]"]
}
},
"rejectedPlans": []
},
"queryShapeHash": "AA3078A59481B28A38775FA280FC4779BEA0E8E900
AF63A4344B76A85859C985",
"serverInfo": {
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"port": 27017,
"version": "8.0.26",
"gitVersion": "a2c24805265db119cc39f2f54e067b4294a4ddd2"
},
"serverParameters": {
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00,
"internalDocumentSourceGroupMaxMemoryBytes": 104857600,

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"internalQueryMaxBlockingSortMemoryUsageBytes": 33554432,
"internalQueryProhibitBlockingMergeOnMongoS": 0,
"internalQueryMaxAddToSetBytes": 104857600,
"internalDocumentSourceSetWindowFieldsMaxMemoryBytes": 104857600,
"internalQueryFrameworkControl": "trySbeRestricted",
"internalQueryPlannerIgnoreIndexWithCollationForRegex": 1
}
}
]

```

Before the index, the query used a **COLLSCAN** (full collection scan): MongoDB examined **all 33 documents** to return **4** matching ones - most of the work was wasted, because only 4 of the 33 documents actually matched the filter.

After the index, the query switched to an **IXSCAN** (index scan): MongoDB examined only **4 documents** (and **4 index keys**) to return the same **4** results. The index let MongoDB jump straight to the matching range instead of reading the whole collection.

Metric	Before	After
Stage	COLLSCAN	IXSCAN
Documents examined	33	4
Keys examined	0	4
Documents returned	4	4
Execution time	0 ms	1 ms

Index reduced the number of examined documents from **33 to 4**, changing the plan from a full collection scan to a targeted index scan. The execution time stayed near zero because the collection is very small, but on a large collection queries would be faster and more scalable.